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Few-Shot Learning Patterns in Financial Time-Series for Trend-Following Strategies

Kieran Wood, Samuel Kessler, Stephen J. Roberts, Stefan Zohren

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Forecasting models for systematic trading strategies do not adapt quickly when financial market conditions rapidly change, as was seen in the advent of the COVID-19 pandemic in 2020, causing many forecasting models to take loss-making positions. To deal with such situations, we propose a novel time-series trend-following forecaster that can quickly adapt to new market

conditions, referred to as regimes. We leverage recent developments from the deep learning community and use few-shot learning. We propose the Cross Attentive Time-Series Trend Network -- X-Trend -- which takes positions attending over a context set of financial time-series regimes. X-Trend transfers trends from similar patterns in the context set to make forecasts, then subsequently takes positions for a new distinct target regime. By quickly adapting to new financial regimes, X-Trend increases Sharpe ratio by 18.9% over a neural forecaster and 10-fold over a conventional Time-series Momentum strategy during the turbulent market period from 2018 to 2023. Our strategy recovers twice as quickly from the COVID-19 drawdown compared to the neural-forecaster. X-Trend can also take zero-shot positions on novel unseen financial assets obtaining a 5-fold Sharpe ratio increase versus a neural time-series trend forecaster over the same period. Furthermore, the cross-attention mechanism allows us to interpret the relationship between forecasts and patterns in the context set.

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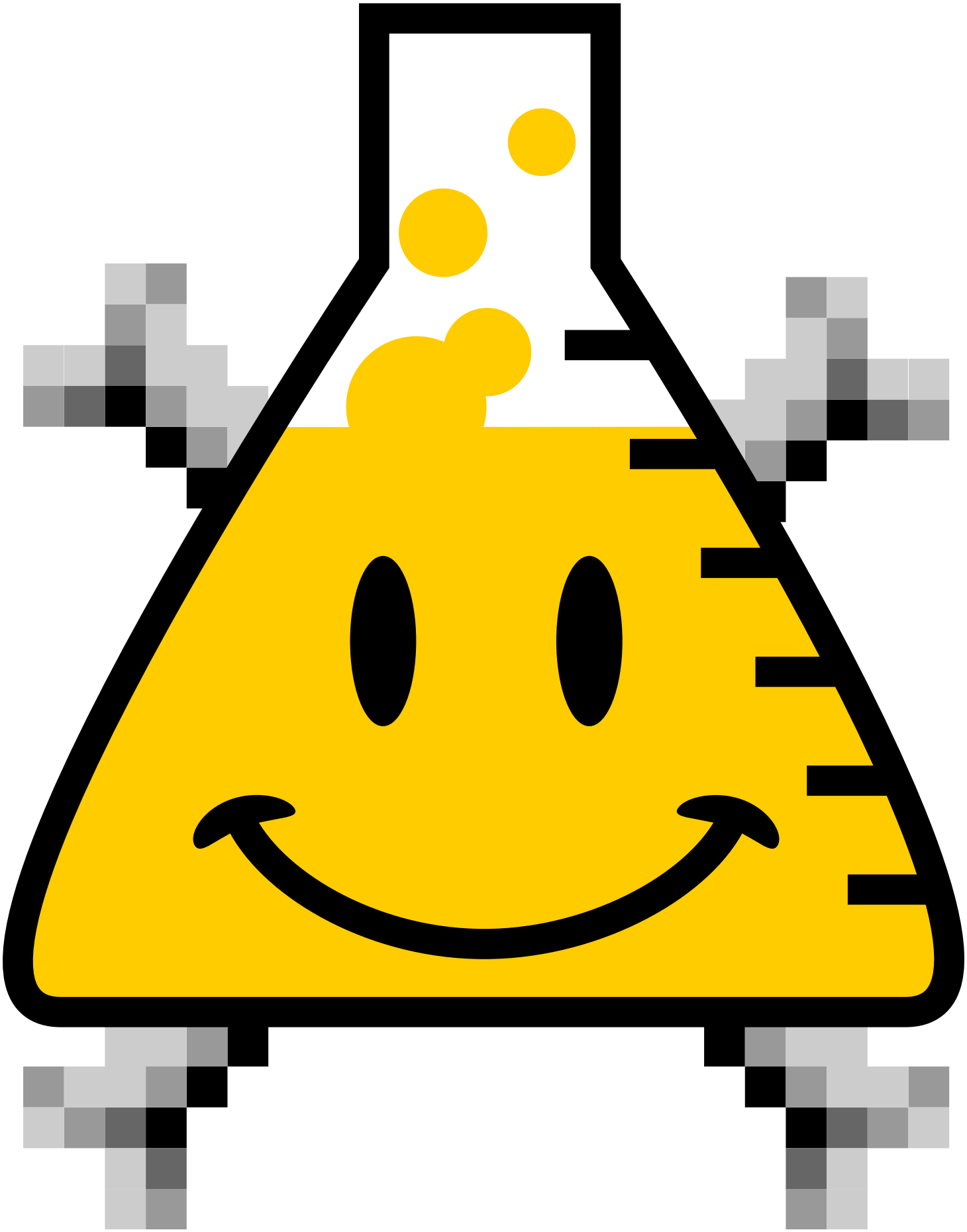

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